

Bayesian Tabulation Audit Boundary Contracts

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Abstract

Bayesian tabulation audits, in the Rivest-Shen tradition, estimate the posterior probability that a full hand count would overturn the reported outcome. Unlike frequentist risk-limiting audits, they report an upset probability under a prior rather than a distribution-free risk limit. This paper documents the RCOUNT V.20 boundary contract for that surface. The implemented verifier does not run a posterior simulation. It verifies that a declared Bayesian audit record is shaped as Bayesian evidence, that prior and likelihood identifiers are present, that declared posterior winner and upset-risk fields are bounded, and that any calibrated risk limit is only a declared field unless a separate calibration rule is implemented. The result is a deliberately narrow contract: posterior analytics can be recorded and replayed as public evidence without being confused with risk-limiting certification.

1 Introduction

Bayesian tabulation audits answer a different question from frequentist risk-limiting audits. In the Rivest-Shen tradition, the audit treats the reported tabulation and the observed evidence as inputs to a posterior model and reports the probability that a full hand count would overturn the reported outcome. That number is useful for explanation and comparison, but it is conditional on the declared prior and likelihood.

RCOUNT therefore records the Bayesian surface as an analytic boundary. The verifier does not accept an exported posterior value as a risk limit, and it does not convert posterior upset probability into frequentist certification. Instead, it checks that a record naming the Bayesian tabulation method carries the fields needed to describe the analysis: prior identity, likelihood identity, posterior winner probability, posterior upset risk, and any declared simulation or calibration metadata.

This paper covers that implemented V.20 slice. It claims only that the Bayesian audit record is well formed and kept separate from risk-limiting replay. It does not claim that RCOUNT currently recomputes a posterior distribution or that a Bayesian upset probability is itself an RLA risk limit.

2 Bayesian Contract

The implemented record is an `AuditAlgorithmRun` whose method id is `bayesian-tabulation-audit-v1`. Its Bayesian fields are explicit and are kept out of non-Bayesian audit methods.

Field	Meaning
<code>method_id</code>	Versioned method name. Current value: <code>bayesian-tabulation-audit-v1</code> .
<code>bayesian_prior_id</code>	Non-empty identifier for the declared prior specification.
<code>bayesian_likelihood_id</code>	Non-empty identifier for the declared likelihood or evidence model.
<code>posterior_winner_probability_ppm</code>	Declared posterior probability, in parts per million, that the reported winner remains the winner.
<code>posterior_risk_ppm</code>	Declared posterior upset probability, in parts per million.
<code>simulation_seed</code>	Optional seed metadata for a posterior computation performed outside this verifier.
<code>posterior_draws</code>	Optional positive count of posterior draws.
<code>calibrated_risk_limit_ppm</code>	Optional declared calibration threshold; the current verifier only checks its shape.
<code>assertions</code>	Non-empty list whose assertion kinds must be <code>bayesian-outcome</code> .

The contract is intentionally a record-shape and design boundary. The prior and likelihood identifiers make the posterior claim attributable. The posterior winner and upset-risk fields are bounded probability fields. The optional calibration field records a possible bridge to risk-limiting language, but it is not treated as a valid frequentist risk limit unless a later calibrated rule is implemented.

3 Implemented Surface

The core check is `verify_bayesian_audit_design` in `crates/rcount-core/src/verify_audit.rs`. It is called from the generic audit-algorithm verifier after method-id support and generic risk-limit shape checks. The method id is the constant `BAYESIAN_TABULATION_AUDIT_METHOD_ID`, whose value is `bayesian-tabulation-audit-v1`.

For non-Bayesian methods, `verify_bayesian_audit_design` rejects any run that carries Bayesian-only fields: prior id, likelihood id, posterior winner probability, posterior risk, simulation seed, posterior draw count, or calibrated risk limit. This prevents a BRAVO, Kaplan-Markov, ALPHA, or other record from smuggling posterior analytics into a risk-limiting method.

For the Bayesian method, the function checks exactly this design boundary: the prior id is present and not blank; the likelihood id is present and not blank; the posterior winner probability is present and no greater than 1,000,000 ppm; the posterior risk is present and no greater than 1,000,000 ppm; an optional posterior draw count is not zero; an optional calibrated risk limit is greater than zero and less than 1,000,000 ppm; at least one assertion is present; and every assertion has kind `bayesian-outcome`. The function does not run a

posterior simulation, recompute the upset probability, or prove calibration to a frequentist risk limit.

The audit crate preserves the same boundary in replay. In `crates/rcount-audit/src/lib.rs`, Bayesian tabulation dispatch returns a boundary transcript with the message that posterior analytics are recorded but are not risk-limiting replay.

Check	Failure caught
<code>audit_algorithm_transcript</code>	Unsupported method ids, duplicate run ids, malformed generic audit records, and Bayesian records that fail the design boundary.
<code>verify_bayesian_audit_design</code>	Missing or blank prior/likelihood ids, missing or out-of-range posterior probability fields, zero posterior draws, invalid calibrated risk threshold, missing assertions, or non-Bayesian assertion kinds.
<code>non-Bayesian Bayesian-field guard</code>	Any non-Bayesian audit method carrying Bayesian-only posterior fields.
<code>algorithm replay boundary</code>	A Bayesian record being reported as posterior analytics rather than as BRAVO, Kaplan-Markov, ALPHA, or another risk-limiting replay.
<code>cvr_summary_total</code>	Public CVR rows and summaries still reconcile before audit records are trusted.
<code>source_hash_match</code>	Source evidence bytes still match the declared source index.

4 Fixtures

The positive fixture is `synthetic_bayesian_tabulation_boundary_package` in `crates/rcount-core/src/fixtures.rs`. It builds on the synthetic summary package and adds one audit algorithm run named `audit-run:bayesian-tabulation-boundary`. The run uses method id `bayesian-tabulation-audit-v1`, sampling mode `boundary-only`, a Dirichlet-multinomial toy prior id, a without-replacement likelihood id, a declared posterior winner probability of 958000 ppm, a declared posterior risk of 42000 ppm, simulation metadata, and one `bayesian-outcome` assertion. Its decision is `boundary`, not `pass`.

The matching core test is `bayesian_tabulation_boundary_package_verifies_analytic_surface`. It verifies the package and checks that the audit-algorithm transcript pass is recorded for `audit-run:bayesian-tabulation-boundary`. The test also asserts the declared posterior winner and risk fields, making the fixture a stable record-shape example rather than an implicit posterior recomputation.

The negative fixture is `synthetic_bad_bayesian_tabulation_boundary_package`. It starts from the positive package and edits `posterior_risk_ppm` to 1000001. That value is outside the one-million-parts-per-million probability range, so `verify_bayesian_audit_design` returns `InvalidBayesianAuditDesign`. The corresponding test is `bayesian_tabulation_boundary_package_rejects_invalid_posterior_risk`.

The audit crate has the parallel replay test `bayesian_algorithm_statistics_reports_documented_boundary`. It runs the positive fixture through `replay_audit_algorithm`

`statistics` and expects an `AlgorithmReplayStatus::Boundary` transcript whose text says the record is not risk-limiting replay.

5 Boundaries

The plan boundary is the governing claim: Bayesian outputs are analytic unless explicitly calibrated to a risk limit. The implementation follows that line. It verifies a declared Bayesian audit record, but it does not perform a full posterior computation and does not certify that the posterior upset probability has frequentist risk-limiting coverage.

Prior dependence is part of the public contract. The same tabulation evidence can produce different posterior upset probabilities under different priors or likelihood models, so the verifier requires non-empty identifiers for both. It still treats those identifiers as declared metadata. A future implementation could bind them to executable posterior code, hand-computable fixtures, or an external calibration proof; V.20 does not yet do that.

Bayesian audits are therefore not risk-limiting in the frequentist sense merely because they report a small upset probability. RCOUNT keeps the field `calibrated_risk_limit_ppm` optional and shape-checked only. Until a calibration rule is added, that field is a boundary marker, not a certificate. This is also why the replay transcript reports Bayesian tabulation as recorded posterior analytics rather than as BRAVO, Kaplan-Markov, ALPHA, or another RLA stopping calculation.

6 Conclusion

RCOUNT V.20 adds a narrow Bayesian tabulation audit boundary. The method id `bayesian-tabulation` is supported, the Bayesian-only fields are excluded from other audit methods, and a Bayesian run must carry non-empty prior and likelihood identifiers, bounded posterior probability fields, a valid optional draw count and calibration threshold, and at least one `bayesian-outcome` assertion.

That is a useful but modest contract. It makes posterior audit analytics public and structurally verifiable, and it prevents those analytics from being misreported as frequentist risk-limiting replay. The remaining work is the work named in the plan: a tiny hand-computable posterior fixture and a calibrated risk boundary fixture. Until then, the implemented surface should be read as a record-shape and reporting-safety contract, not as a full Bayesian audit engine.

References

- Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.
- Gio Della-Libera. Bayesian tabulation audit boundary contracts. Technical report, Apportionment Research, 2026b.